

Практикум D6

114 marks of the archive, six techniques, twelve tasks. Everything the archive asks about a random variable given by a probability density function, from May 2021 to November 2025: an area, the constant that makes the whole area one, the median and the quartiles, the mode, the mean and the variance, and a second question around an area.

О практикуме

Плотностная половина

statistics.continuous_random_variables: 90 баллов из 199 (0.452) — всё, где величина задана формулой, а не кривой с μ и σ . 10 баллов statistics.expected_value (0.167 — 10 из 60): ожидаемая сумма покупки и E , Var по плотности. 14 баллов statistics.probability (0.059 — 14 из 237): три условные под плотностью и доля времени в синусоидальной модели. 114 баллов, 35 блоков, и дубля нет — впервые за шесть практикумов статистики. 95 баллов на Paper 2, 19 на Paper 1; из 95 калькуляторных 29 — алгебра «show that», точное $E(X)$ и формулы через n .

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-d6-density

Собрано из ноутбука репозитория `ib-math-aa-hl-past-papers`. Проверки в PDF не работают — они для интерактивной версии.

D6 — Continuous random variables: the density does everything

114 marks of the archive, six techniques, twelve tasks. Everything the archive asks about a random variable given by a probability density function, from May 2021 to November 2025: an area, the constant that makes the whole area one, the median and the quartiles, the mode, the mean and the variance, and a second question around an area.

Almost all of it is on Paper 2 — and a quarter of that is algebra the calculator never touches: *show that* from $\int f = 1$, formulas in n , an exact $E(X)$.

The one idea

A continuous random variable has no table. It has a **density** f , zero outside the interval the question gives, and

a probability is the area under f , above the values where the event happens.

$$P(a < X < b) = \int_a^b f(x) \, dx$$

A single value has no area, so $P(X \leq a) = P(X < a)$.

And what follows from it

Every other question is the same area, used differently:

the question asks	what it is
the constant k	the whole area is 1
the median, a quartile	the boundary with $\frac{1}{2}$, $\frac{1}{4}$ of the area to its left
the mode	where f is highest — not an area at all
$E(X)$, $\text{Var}(X)$	the areas under $x f(x)$ and $x^2 f(x)$

How the checks work

They do not know the answers. Each check is handed **the density** and the facts the question gives:

```
X = Density({(0, 4): x/sqrt((x**2 + k)**3)},  
'X')  
verify_letters('3b', 0.645, k, [X])
```

is *"your k : is the area under f one?"* — the check adds $\int f = 1$ by itself, finds k , and compares. `Density({interval: formula})` is the density exactly as the question prints it, *"0 otherwise"* included.

When you are wrong the check says **how**:

what you wrote	what the check says
the area on the other side	the opposite event
$\int_0^m$ when f starts at 0.5	the formula does not apply left of 0.5
a root outside the interval	that root is rejected
$E(X^2)$ for the variance	the square of the mean is not subtracted
$f(1.5) = 0.441$ for the mode	the mode is the x , not the height
1.65 on Paper 1	the question asks for the exact value

Order of work

level	what it means	tasks
	the area, and the constant that makes it one	1-3
	the median and quartiles, the mode	4-8
	the mean and the variance; a second question around an area	9-12

Every task is a real past-paper question, cited.

95 of these 114 marks are on a calculator paper, and the number

flatters. 29 of the 95 are algebra: *show that*, $E(X)$ in terms of n , an exact value. The calculator really works on the other 66 — an integral nobody does by hand, a median equation with arccos in it.

```
import sys
sys.path.append('.')          # from
practicum/statistics to practicum/kit/
import sympy as sp            # the escape
hatch: anything not in kit is in sp
from kit import *              # checks +
Density, P(), Expect, Var, SD

language('en')                # this
notebook is in English, and so are the checks

# Density({interval: formula}, name) is f,
# and 0 outside the intervals.
# Comparisons are events, P() finds their
# area; & is "and", | is "or".
#     X = Density({(0, 2): 3*x**2/8}, 'X')
#     P(X < 1.5), P((X > 0.5) & (X < 1.5)),
#     P(X > 1.5, given=X > 1)
# Expect(X), Var(X), SD(X) are integrals.
# Letters are allowed: the checks find them.

print('ready; sympy', sp.__version__)
X = Density({(0, 2): 3*x**2/8}, 'X')
print('the model:      ', X)
print('P(X < 1.5):     ', sympify(P(X <
1.5)))
print('E(X):          ', sympify(Expect(X)))
```

Map of the six techniques

#	technique	you recognise it by	it reduces to
1	the area	<i>f</i> fully known; <i>find the probability, at most \$48, a random time</i>	$\int f$ over the right interval
2	the constant	<i>k, a, b</i> in <i>f</i> and nothing else said; <i>show that $324a + 72b = 1$</i>	$\int f = 1$ over the whole interval
3	the median, a quartile	<i>find the median, the lower quartile, $P(X - m \leq a) = 0.3$</i>	$\int_{\text{start}}^m f = \frac{1}{2}$
4	the mode	<i>find the mode, which is greater, the mode or the median</i>	the maximum of <i>f</i>
5	the mean and the variance	$E(X)$, $\text{Var}(X)$, <i>the expected amount spent</i>	$\int xf$, $\int x^2 f - \mu^2$
6	a second question	<i>given that, $P(\mu - \sigma < X < \mu + \sigma)$</i>	a ratio of areas; moments first

Technique 1 is the whole topic read forwards. Techniques 2 and 3 read it backwards: the area is known, a letter is not. Technique 4 is the one place where the answer is not an area. Techniques 5 and 6 integrate something else, or put a question around the area.

Part 1. The area, and the constant that makes it one

Theory: an area under a formula

Take $f(x) = \frac{3}{8}x^2$ for $0 \leq x \leq 2$, and 0 otherwise.

Draw it first, and shade what the question names. Then integrate over the shaded interval — and only where f is given:

$$\begin{aligned}P(X < 1.5) &= \int_0^{1.5} \frac{3}{8}x^2 \, dx \\&= \left[\frac{1}{8}x^3 \right]_0^{1.5} \\&= \frac{27}{64} \\&= 0.422\end{aligned}$$

$P(X < 3)$ is 1, not $\frac{27}{8}$: beyond 2 the density is zero, and the formula no longer applies.

On a calculator paper the integral is `fnInt`; the method mark is for writing

$\int_0^{1.5} f(x) dx$ with the right limits.

A piecewise density is integrated piece by piece: the area up to a point on the second piece is the whole first piece plus part of the second.

An event about another quantity is translated into an interval of X first. Postage on a parcel of X kg is \$4 per kg below 1.5 kg and \$3 per kg from 1.5 kg. "*Postage at most \$5*" is $4X \leq 5$ on the first rate, so $X \leq 1.25$; **and** $3X \leq 5$ on the second, so $1.5 \leq X \leq \frac{5}{3}$. Two intervals — and $X \leq \frac{5}{4}$ alone loses the second.

A random time is the simplest density of all: a moment chosen at random in the first T seconds has $f(t) = \frac{1}{T}$, and the probability of anything is the fraction of the time it holds.

Task 1 ● — **May 2023 TZ2 Paper 2**
Q10(e), 4 marks

A weight suspended on a spring is pulled down and released, so that it moves up and down vertically.

The height, H metres, of the base of the weight above the ground can be modelled by the function $H(t) = a \cos(7.8t) + b$, for $a, b \in \mathbb{R}$ and $0 \leq t \leq 10$, where t is the time in seconds after the weight is released.

The weight is released when its base is at a minimum height of 1 metre above the ground, and it reaches a maximum height of 1.8 metres above the ground. In part (b) this gives $a = -0.4$ and $b = 1.4$.

A camera is set to take a picture of the weight at a random time during the first five seconds of its motion.

(e) Find the probability that the height of the base of the weight is greater than 1.5 metres at the time the picture is taken.

Parts (a)–(d) are a sinusoidal model — C4. The check does not know where $H > 1.5$: it walks along $[0, 5]$ and finds where it starts and stops.

```

qle = ...          # P(H > 1.5) at a random
time in the first five seconds

T = Density(({(0, 5): Rational(1, 5)}), 'T',
var=t)
H = T.map(lambda s: -0.4*cos(7.8*s) + 1.4,
'H')

verify_chance('le', qle, P(H > 1.5))

```

Theory: the whole area is one

A density with a letter in it has exactly one fact about the letter that the question need not state: **the area under f is one.**

$f(x) = k(4 - x^2)$ for $0 \leq x \leq 2$:

$$\begin{aligned}
 & \int_0^2 k(4 - x^2) \\
 dx &= k \left[4x - \frac{1}{3}x^3 \right]_0^2 = \frac{16}{3}k = 1 \\
 & \implies \\
 k &= \frac{3}{16}
 \end{aligned}$$

Write " $= 1$ ". The markscheme gives a method mark for it, and a *show that* without it collapses: the working proves nothing.

The integral is the work. Substitution for $\frac{x}{(x^2+c)^{3/2}}$ or $\frac{c}{\sqrt{1-x^2}}$ (arcsin appears), parts for xe^x , piece by piece when the density is piecewise — and then the limits, which may themselves be letters: for $f(x) = cx$ on $[0, c]$ the area is $\frac{c^3}{2}$.

Two letters need two facts. $f(x) = px + q$ on $[0, 2]$ with $f(2) = 0$: the area gives $2p + 2q = 1$, the value gives $2p + q = 0$, so $q = 1$ and $p = -\frac{1}{2}$. One equation for two letters has infinitely many answers.

Paper 1 wants the exact value. $\frac{3}{16}$, $\frac{2}{\pi}$
— not 0.1875, not 0.637.

Task 2 — **May 2022 TZ1 Paper 1 Q7(a), 4 marks**

The continuous random variable X has probability density function

$$f(x) = \begin{cases} \frac{k}{\sqrt{4-3x^2}}, & 0 \leq x \leq 1 \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the value of k .

Paper 1: the exact value. Part (b), $E(X)$, is in the archive.

```

q2a = ...          # k, exact

k = symbols('k', positive=True)
X = Density({(0, 1): k/sqrt(4 - 3*x**2)},
            'X')

verify_letters('2a', q2a, k, [X], exact=True)

```

Task 3 — May 2021 TZ1 Paper 2 Q7, 7 marks

A continuous random variable X has the probability density function f given by

$$f(x) = \begin{cases} \frac{x}{\sqrt{(x^2 + k)^3}}, & 0 \leq x \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

where $k \in \mathbb{R}^+$.

(a) Show that $\sqrt{16 + k} - \sqrt{k} = \sqrt{k} \sqrt{16 + k}$.

(b) Find the value of k .

For (a) enter the area under f as an expression in k — the line just before you set it equal to 1. The check tries several k

.

```
k = symbols('k', positive=True)

q3a = ...          # the area under f, in terms
of k
q3b = ...          # k

X = Density({(0, 4): x/sqrt((x**2 + k)**3)},
            'X')

verify_chance('3a', q3a,
total_probability(X), free=k)
verify_letters('3b', q3b, k, [X])
```

Part 2. The median, the quartiles, the mode

Theory: a boundary from a share of the area

The median m has half the area to its left; the lower quartile a quarter; the upper quartile three quarters.

$$\int_{\text{start}}^m f(x) \, dx = \frac{1}{2}$$

For $f(x) = \frac{3}{8}x^2$ on $[0, 2]$: $\frac{1}{8}m^3 = \frac{1}{2}$, so $m = \sqrt[3]{4} = 1.59$; the lower quartile is $\sqrt[3]{2} = 1.26$.

The lower limit is where f starts. If f lives on $[0.8, 3]$, the median equation starts at 0.8 — integrating the formula from 0 adds an area that is not there.

Piecewise: find the piece first. $f(x) = \frac{x}{2}$ on $[0, 1]$ and $f(x) = \frac{1}{2}$ on $[1, 2.5]$. The first piece holds $\frac{1}{4}$ — less than a half, so the median is on the second piece:

$$\frac{1}{4} + \frac{1}{2}(m - 1) = \frac{1}{2} \implies m = 1.5$$

Using the first formula everywhere, $\frac{m^2}{4} = \frac{1}{2}$, gives $\sqrt{2}$ — a number, but not the median.

Keep only the root on its piece. A quadratic median equation has two roots, and one of them lies outside the interval: the markscheme gives a mark for rejecting it.

Around the median. $P(|X - m| \leq d)$ is the area from $m - d$ to $m + d$ — one more equation, now for d .

Task 4 — **November 2022 Paper 2 Q6, 8 marks**

The continuous random variable X has a probability density function given by

$$f(x) = \begin{cases} axe^x, & 0 \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$$

where $a, b \in \mathbb{R}^+$.

- (a)** Find an expression for a in terms of b .
- (b)** In the case where $a = b = 1$, find the median of X .

In (a) the answer contains b: the check tries several b.

```
a, b = symbols('a b', positive=True)
m = symbols('m')

q4a = ...          # a in terms of b
q4b = ...          # the median when a = b = 1

X = Density({(0, b): a*x*exp(x)}, 'X')
X1 = Density({(0, 1): x*exp(x)}, 'X')

verify_letters('4a', q4a, a, [X], free=b)
verify_letters('4b', q4b, m, [X1], [Eq(P(X1 <
m), 0.5)])
```

Task 5 — **May 2024 TZ1 Paper 2 Q8, 6 marks**

A continuous random variable X has a probability density function f given by

$$f(x) = \begin{cases} kx, & 0 \leq x \leq k \\ 2kx - x^2, & k < x \leq 2k \\ 0, & \text{otherwise} \end{cases}$$

where $k > 0$.

(a) Show that k satisfies the equation $7k^3 = 6$.

(b) Find the median of X .

For (a) enter the area under f in terms of k . In (b) the check finds k itself from the area being one.

```
k = symbols('k', positive=True)
m = symbols('m')

q5a = ...          # the area under f, in terms
of k
q5b = ...          # the median

X = Density(({(0, k): k*x, (k, 2*k): 2*k*x -
x**2}), 'X')

verify_chance('5a', q5a,
total_probability(X), free=k)
verify_letters('5b', q5b, m, [X], [Eq(P(X <
m), 0.5)])
```

Task 6 — **November 2021 Paper 2 Q7, 6 marks**

A continuous random variable X has a probability density function given by

$$f(x) = \begin{cases} \arccos x, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

The median of this distribution is m .

(a) Determine the value of m .

(b) Given that $P(|X - m| \leq a) = 0.3$, determine the value of a .

The check for (b) finds m itself, so a slip in (a) costs (a) only.

```
q6a = ...          # m
q6b = ...          # a

m = symbols('m')
a = symbols('a', positive=True)
X = Density({(0, 1): acos(x)}, 'X')
median = Eq(P(X < m), 0.5)

verify_letters('6a', q6a, m, [X], [median])
verify_letters('6b', q6b, a, [X], [median,
Eq(P((X >= m - a) & (X <= m + a)), 0.3)])
```

Task 7 — **May 2022 TZ2 Paper 1** **Q8, 6 marks**

A continuous random variable X has the probability density function

$$f(x) = \begin{cases} \frac{2}{(b-a)(c-a)}(x-a), & a \leq x \leq c \\ \frac{2}{(b-a)(b-c)}(b-x), & c < x \leq b \\ 0, & \text{otherwise} \end{cases}$$

The graph of $y = f(x)$ for $a \leq x \leq b$ is a triangle: it rises from 0 at $x = a$ to its

peak at $x = c$ and falls back to 0 at $x = b$.

Given that $c \geq \frac{a+b}{2}$, find an expression for the median of X in terms of a , b and c .

Paper 1. The check tries several triangles with $c \geq \frac{a+b}{2}$ and compares your expression with the median of each.

```
a, b, c, m = symbols('a b c m')

q7 = ...          # the median in terms of a,
b, c

X = Density({(a, c): 2*(x - a)/((b - a)*(c -
a)),
              (c, b): 2*(b - x)/((b - a)*(b -
c))}, 'X')
triangles = [{a: 0, b: 4, c: 3}, {a: 1, b: 3,
c: Rational(5, 2)}, {a: -2, b: 6, c: 5}]

verify_letters('7', q7, m, [X], [Eq(P(X < m),
0.5)], free=triangles)
```

Theory: the mode is a peak, not an area

The mode is the value of x where f is **largest**. Find it from the graph, from $f'(x) = 0$, or from the axis of symmetry of

a parabola — and remember the ends of the interval.

- $f(x) = \frac{3}{8}x^2$ on $[0, 2]$ increases: the mode is at the end, $x = 2$.
- $f(x) = 12x^2(1 - x)$ on $[0, 1]$: $f'(x) = 24x - 36x^2 = 0$ at $x = \frac{2}{3}$.

The answer is the x . $f(\frac{2}{3}) = \frac{16}{9}$ is the height of the peak, and the markscheme gives no accuracy mark for it.

Mode against median — without the median. Compute the area to the left of the mode. For $12x^2(1 - x)$ it is $4x^3 - 3x^4$ at $\frac{2}{3}$, which is $\frac{16}{27} > \frac{1}{2}$: half the area is reached **before** the mode, so the median is less than the mode. That one inequality is the reason the markscheme's R1 asks for.

Task 8 — **May 2025 TZ1 Paper 2 Q11(a), (b), (d), 9 marks**

In a marathon race, the random variable T represents the time, in hours, taken for a runner to complete the race. No runner completes the race in less than 2.25

hours, and no runner completes it in more than 7.5 hours.

The probability density function for T is modelled by f , defined by

$$f(t) = \begin{cases} \frac{4}{21} \left(1 - \cos \left(\frac{4\pi}{9}(t - 2.25) \right) \right), & 2.25 \leq t < 4.5 \\ \frac{4}{21} \left(1 + \cos \left(\frac{\pi}{3}(t - 4.5) \right) \right), & 4.5 \leq t \leq 7.5 \\ 0, & \text{otherwise.} \end{cases}$$

The graph of f has a maximum point at $t = 4.5$.

(a) (i) Find the value of $\int_{2.25}^{4.5} f(t) dt$. **(ii)** Write down the mode of T . **(iii)** Determine which is greater, the mode of T or the median of T , justifying your answer.

The runners who finish the race in 3.5 hours or less are considered to be fast runners.

(b) Find the probability that a runner chosen at random is a fast runner.

(d) Find the lower quartile of T .

For (a)(iii) answer 'mode' or 'median'. Part (c) is technique 6 — it is in the archive; (e) and (f) are $E(a - bT)$ — D4.

```
q8ai = ...      # the integral from 2.25 to
4.5
q8aai = ...     # the mode
q8aiii = ...    # 'mode' or 'median' — which
is greater
q8b = ...       # P(fast runner)
q8d = ...       # the lower quartile

Q1 = symbols('Q1')
T = Density({(Rational(9, 4), Rational(9,
2)): Rational(4, 21)*(1 - cos(4*pi/9*(t -
Rational(9, 4)))),
              (Rational(9, 2), Rational(15,
2)): Rational(4, 21)*(1 + cos(pi/3*(t -
Rational(9, 2))))},
            'T', var=t)

verify_chance('8a(i)', q8ai, P(T < 4.5))
verify_mode('8a(ii)', q8aai, T)
verify_greater('8a(iii)', q8aiii, T)
verify_chance('8b', q8b, P(T <= 3.5))
verify_letters('8d', q8d, Q1, [T], [Eq(P(T <
Q1), 0.25)])
```

Part 3. The mean, the variance, and a second question

Theory: the mean and the variance are integrals too

In D4 the mean was $\sum x P(X = x)$. With a density the sum becomes an integral, over the interval where f lives:

$$\begin{aligned} E(X) &= \int x \\ &\quad f(x) \\ &\quad dx, \\ \text{Var}(X) &= \int x^2 f(x) \\ &\quad dx - (E(X))^2 \end{aligned}$$

For $f(x) = \frac{3}{8}x^2$ on $[0, 2]$:

$$\begin{aligned} E(X) &= \int_0^2 \frac{3}{8} x^3 \\ dx &= 1.5, \\ E(X^2) &= \int_0^2 \frac{3}{8} x^4 \\ dx &= 2.4, \\ \text{Var}(X) &= 2.4 - 1.5^2 = 0.15 \end{aligned}$$

$E(X^2)$ is not the variance. 2.4 is the most common wrong answer here, and a markscheme gives it (M1)A0A0A0.

Letters stay letters. For $f(x) = \frac{1}{3c}$ on $[c, 4c]$ the mean is $\frac{5c}{2}$ by symmetry, and $E(X^2) = \int_c^{4c} \frac{x^2}{3c} dx = 7c^2$.

An exact value needs the substitution:
 $\int \frac{x}{\sqrt{9-x^2}} dx = -\sqrt{9-x^2}.$

The mean of something else. $E(g(X)) = \int g(x) f(x) dx$. The postage from Theory 1 is $4x$ below 1.5 kg and $3x$ from 1.5 kg, so the expected postage is two integrals, one for each rate:

$$\int_{\text{start}}^{1.5} 4x f(x) dx + \int_{1.5}^{\text{end}} 3x f(x) dx$$

Multiplying $E(X)$ by one rate is wrong on both pieces.

Task 9 — *May 2023 TZ2 Paper 1 Q6, 5 marks*

A continuous random variable X has probability density function f defined by

$$f(x) = \begin{cases} \frac{1}{2a}, & a \leq x \leq 3a \\ 0, & \text{otherwise} \end{cases}$$

where a is a positive real number.

(a) State $E(X)$ in terms of a .

(b) Use integration to find $\text{Var}(X)$ in terms of a .

The checks try several values of a .

```
a = symbols('a', positive=True)

q9a = ...          # E(X) in terms of a
q9b = ...          # Var(X) in terms of a

X = Density({(a, 3*a): 1/(2*a)}, 'X')

verify_moment('9a', q9a, Expect(X))
verify_moment('9b', q9b, Var(X))
```

Theory: a second question around an area

Given that. The condition is the new whole; the event is the part of it that also happens. For $f(x) = \frac{3}{8}x^2$ on $[0, 2]$:

$$\begin{aligned}P(X > 1.5 \mid X > 1) &= \frac{P(X > 1.5)}{P(X > 1)} \\&= \frac{1 - \frac{27}{64}}{1 - \frac{1}{8}} \\&= 0.661\end{aligned}$$

Here the event lies inside the condition, so the overlap is the event itself. When it does not, the numerator is the area of the overlap.

An interval built from the moments.

$P(\mu - \sigma < X < \mu + \sigma)$ needs the moments first: $\mu = 1.5$, $\sigma = \sqrt{0.15} = 0.387$, and then the area from 1.113 to 1.887 is 0.668.

σ , not σ^2 . The interval is one *standard deviation* each side; $\mu \pm 0.15$ is a different, narrower question.

Task 10 ● — **November 2025 TZ1**
Paper 2 Q11(c), (d), 8 marks

The probability density function of a random variable X is given by

$$f(x) = \begin{cases} 3x \arccos(x^2), & 0 \leq x \leq k \\ 0, & \text{otherwise.} \end{cases}$$

In part (b), k is found from $k^2 \arccos(k^2) - \sqrt{1 - k^4} + \frac{1}{3} = 0$.

(c) Find **(i)** $E(X)$; **(ii)** $\text{Var}(X)$.

(d) Given that $\mu = E(X)$ and $\sigma^2 = \text{Var}(X)$, find $P(\mu - \sigma < X < \mu + \sigma)$.

Parts (a) and (b) are integration — E5 and E6. The checks find k themselves from the area being one.

```

q10ci = ...      # E(X)
q10cii = ...     # Var(X)
q10d = ...       # P(mu - sigma < X < mu +
sigma)

k = symbols('k', positive=True)
X = Density({(0, k): 3*x*acos(x**2)}, 'X')
mu, sigma = Expect(X), SD(X)

verify_moment('10c(i)', q10ci, Expect(X),
given=[], var=k)
verify_moment('10c(ii)', q10cii, Var(X),
given=[], var=k)
verify_chance('10d', q10d, P((X > mu - sigma)
& (X < mu + sigma)), given=[], var=k)

```

Task 11 — **May 2024 TZ2 Paper 2** **Q10(d), (e), 8 marks**

A shop sells chocolates. The weight, in kilograms, of chocolates bought by a random customer can be modelled by a continuous random variable X with probability density function f defined by

$$f(x) = \begin{cases} \frac{6}{85} (4 + 3x - x^2), & 0.5 \leq x \leq 3 \\ 0, & \text{otherwise.} \end{cases}$$

The shop sells chocolates to customers at \$25 per kilogram.

However, if the weight of chocolate bought by a customer is at least 0.75 kilograms, the shop sells chocolate at a discounted rate of \$24 per kilogram.

(d) Find the probability that a randomly selected customer spends at most \$48.

(e) Find the expected amount spent per customer. Give your answer correct to the nearest cent.

Parts (a)–(c) — the mode, an area, the median — are in the archive.

```
q11d = ...          # P(spend <= 48)
q11e = ...          # the expected amount spent,
to the nearest cent

X = Density({(Rational(1, 2), 3): Rational(6,
85)*(4 + 3*x - x**2)}, 'X')
spend = X.map(lambda w: Piecewise((25*w, w <
0.75), (24*w, True)), 'spend')

verify_chance('11d', q11d, P(spend <= 48))
verify_moment('11e', q11e, Expect(spend),
places=2)
```

Task 12 ● — November 2025 TZ3
Paper 2 Q11(a), (b)(ii), (c), (d), 15
marks

The lengths, in metres, of jumps in a long jump competition can be modelled by a continuous random variable X with probability density function f defined by:

$$f(x) = \begin{cases} 0, & x < 3 \\ a(x-3)^3 + b(x-3)^2, & 3 \leq x \leq 9 \\ 0, & x > 9 \end{cases}$$

where $a, b \in \mathbb{R}$, $a, b \neq 0$.

(a) Show that $324a + 72b = 1$.

It is given that $f(9) = 0$. In part (b)(i) this becomes $6a + b = 0$.

(b) (ii) Determine the value of a and the value of b .

(c) Show that the median of X is less than the mode of X .

Matt has the final jump in the competition. He needs to jump at least 8.52 m to win the competition.

(d) Given that he jumps over 8 m, use the model to find the probability that he wins the competition.

For (a) enter the area under f in terms of a and b . For (c) enter the mode and the median; the check also wants the median to three figures. The checks for (c) and (d) find a and b themselves.

```
a, b, m = symbols('a b m')

q12a = ...           # the area under f, in
terms of a and b
q12bii = [...]       # [a, b]
q12c_mode = ...      # the mode
q12c_median = ...    # the median
q12d = ...           #  $P(X > 8.52 \mid X > 8)$ 

X = Density(({(3, 9): a*(x - 3)**3 + b*(x -
3)**2}, 'X'))
zero = [Eq(6*a + b, 0)]

verify_chance('12a', q12a,
total_probability(X), free=[a, b])
verify_letters('12b(ii)', q12bii, [a, b],
[X], zero)
verify_mode('12c mode', q12c_mode, X,
given=zero, var=[a, b])
verify_letters('12c median', q12c_median, m,
[X], zero + [Eq(P(X < m), 0.5)])
verify_chance('12d', q12d, P(X > 8.52,
given=X > 8), given=zero, var=[a, b])
```

Trainer: name the technique in five seconds

Twelve openings. Do not compute anything — say only **which move you would make first**.

code	technique
area	the density is known: find an area
constant	a letter in f , found from the whole area being one
quantile	a median or a quartile: a share of the area
mode	the mode, or the mode against the median
moments	the mean or the variance
condition	given that, or an interval built from the moments

1. $f(x) = cx^3$ for $0 \leq x \leq 2$, and 0 otherwise. Find c .
2. $f(x) = \frac{1}{2} \sin x$ for $0 \leq x \leq \pi$. Find the probability that X is greater than $\frac{\pi}{3}$.
3. $f(x) = 2x$ for $0 \leq x \leq 1$. Find the value of x below which half of the distribution lies.

4. $f(x) = \frac{3}{4}(1 - x^2)$ for $-1 \leq x \leq 1$. Write down the most likely value of X .
5. $f(x) = \frac{2}{x^2}$ for $1 \leq x \leq 2$. Find $E(X)$.
6. $f(x) = \frac{1}{8}x$ for $0 \leq x \leq 4$. Given that $X > 1$, find the probability that $X > 3$.
7. A bus arrives at a random moment in the next 12 minutes. Find the probability that you wait more than 8 minutes.
8. $f(x) = p + qx$ for $0 \leq x \leq 4$, with $f(4) = 0$. Find p and q .
9. $f(x) = \frac{1}{9}x^2$ for $0 \leq x \leq 3$. Find the upper quartile.
0. $f(x) = e^{-x}$ for $x \geq 0$. Find $\text{Var}(X)$.
1. $f(x) = xe^{-x}$ for $x \geq 0$. Show that the median is greater than the mode.
2. $f(x) = \frac{3}{2}x^2$ for $-1 \leq x \leq 1$. Find $P(\mu - \sigma < X < \mu + \sigma)$.

```

answers = {
    1: '', 2: '', 3: '', 4: '', 5: '', 6: '',
    7: '', 8: '', 9: '', 10: '', 11: '', 12:
    '',
}

trigger_check(answers, {1: '96f6549c6b97', 2:
'4a91ee5f0106', 3: 'de00222d4f99', 4:
'e642b12901a6', 5: '690a8d18f75d', 6:
'596d40a4ed24', 7: '4a91ee5f0106', 8:
'96f6549c6b97', 9: 'de00222d4f99', 10:
'690a8d18f75d', 11: 'e642b12901a6', 12:
'596d40a4ed24'})

```

On the clock — *May 2023 TZ1* *Paper 2 Q11(a), (b), 7 marks*

Seven marks, nine minutes.

Calculator allowed, no hints.

A continuous random variable, X , has a probability density function defined by

$$f(x) = \begin{cases} \frac{6}{\pi\sqrt{16-x^2}}, & 0 \leq x \leq 2 \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the exact value of $E(X)$.

(b) Find $P(X < 0.5)$.

Attempt log

date	time	result

```
qt_a = ...      # E(X), exact
qt_b = ...      # P(X < 0.5)

X = Density({(0, 2): 6/(pi*sqrt(16 - x**2))},
            'X')

verify_moment('timer (a)', qt_a, Expect(X),
              exact=True)
verify_chance('timer (b)', qt_b, P(X < 0.5))
```



Solutions

Work these only after you have your own answer, or you are reading, not practising.

1 (e) $H(t) = -0.4 \cos(7.8t) + 1.4$ starts at its minimum. It first reaches 1.5 at $t = 0.233779\dots$ and comes back down through 1.5 at $t = 0.571757\dots$, so in each period it is above 1.5 for $0.337978\dots$ s. The period is $\frac{2\pi}{7.8} = 0.805536\dots$ s, and the first five seconds hold six full periods; in the unfinished seventh the weight does not get back above 1.5 before $t = 5$. So

$$\begin{aligned} P(H > 1.5) &= \frac{6 \times 0.337978\dots}{5} \\ &= \frac{2.02787\dots}{5} \\ &= 0.405574\dots \\ &= \boxed{0.406} \end{aligned}$$

2 (a) $\int \frac{k}{\sqrt{4-3x^2}} dx = \frac{k}{\sqrt{3}} \arcsin\left(\frac{\sqrt{3}}{2}x\right)$, and between 0 and 1 that is $\frac{k}{\sqrt{3}} \cdot \frac{\pi}{3}$. Setting it equal to 1:

$$k = \frac{3\sqrt{3}}{\pi} \quad \boxed{k = \frac{3\sqrt{3}}{\pi}}$$

3 (a) With $u = x^2 + k$: $\int \frac{x}{(x^2+k)^{3/2}} dx = -(x^2+k)^{-1/2}$, so the area is

$$\boxed{\frac{1}{\sqrt{k}} - \frac{1}{\sqrt{16+k}}} = 1$$

and multiplying by $\sqrt{k}\sqrt{16+k}$ gives $\sqrt{16+k} - \sqrt{k} = \sqrt{k}\sqrt{16+k}$.

3 (b) Solving on the calculator: $k = 0.645038\dots = \boxed{0.645}$.

4 (a) By parts, $\int_0^b axe^x dx = a[xe^x - e^x]_0^b = a(be^b - e^b + 1) = 1$:

$$\boxed{a = \frac{1}{be^b - e^b + 1}}$$

4 (b) With $a = b = 1$: $\int_0^m x e^x dx = me^m - e^m + 1 = \frac{1}{2}$, so $m = 0.768039\dots = \boxed{0.768}$.

5 (a) $\int_0^k kx \, dx + \int_k^{2k} (2kx - x^2) \, dx = \frac{k^3}{2} + (3k^3 - \frac{7}{3}k^3) = \boxed{\frac{7k^3}{6}}$, and this is 1, so $7k^3 = 6$.

5 (b) $k = 0.949914\dots$, and the first piece holds $\frac{k^3}{2} = \frac{3}{7} < \frac{1}{2}$, so the median is on the second piece:

$$\begin{aligned} \frac{3}{7} + \left[kx^2 - \frac{1}{3}x^3 \right]_k^m &= \frac{1}{2} \\ \implies m &= 1.02925\dots = \boxed{1.03} \end{aligned}$$

The markscheme warns that $\int_0^m kx \, dx = 0.5$ happens to give 1.03 as well — but $m > k$, the first formula does not apply there, and that route scores M1A0M0A0. No check on a number can see the difference.

6 (a) $\int_0^m \arccos x \, dx = m \arccos m - \sqrt{1-m^2} + 1 = 0.5$, so $m = 0.360034\dots = \boxed{0.360}$.

6 (b) $P(m-a \leq X \leq m+a) = \int_{m-a}^{m+a} \arccos x \, dx = 0.3$, and the calculator

gives $a = 0.124861 \dots = \boxed{0.125}$.

7 Since $c \geq \frac{a+b}{2}$, the median is on the rising side. The area up to m is a triangle with base $m - a$ and height $f(m)$:

$$\begin{aligned} \frac{1}{2}(m - a) \cdot \frac{2(m - a)}{(b - a)(c - a)} &= \frac{(m - a)^2}{(b - a)(c - a)} \\ &= \frac{1}{2} \end{aligned}$$

so $m = a \pm \sqrt{\frac{(b-a)(c-a)}{2}}$, and $m \geq a$ rejects the minus sign:

$$\boxed{m = a + \sqrt{\frac{(b - a)(c - a)}{2}}}$$

8 (a)(i) $\int_{2.25}^{4.5} f(t) dt = \frac{4}{21} \times 2.25 = \boxed{\frac{3}{7}} = 0.429$

— the cosine integrates to zero over the half wave.

8 (a)(ii) The maximum is at $t = 4.5$: mode $\boxed{4.5}$.

8 (a)(iii) $P(T < 4.5) = \frac{3}{7} < 0.5$: half the area is reached only after 4.5, so the

median is greater (4.69). median

8 (b) $\int_{2.25}^{3.5} f(t) dt = 0.103749 \dots = \boxed{0.104}$

8 (d) $\int_{2.25}^q f(t) dt = 0.25$ on the first piece (it holds $\frac{3}{7} > \frac{1}{4}$), so $q = 4.01290 \dots = \boxed{4.01}$.

9 (a) The density is constant on $[a, 3a]$: by symmetry $E(X) = \boxed{2a}$.

9 (b) $E(X^2) = \int_a^{3a} \frac{x^2}{2a} dx = \left[\frac{x^3}{6a} \right]_a^{3a} = \frac{13a^2}{3}$, so

$$\text{Var}(X) = \frac{13a^2}{3} - (2a)^2 = \boxed{\frac{a^2}{3}}$$

10 (c) With $k = 0.713250$: $E(X) = \int_0^k 3x^2 \arccos(x^2) dx = 0.456309 \dots = \boxed{0.456}$;
 $E(X^2) = 0.237198 \dots$, so $\text{Var}(X) = 0.237198 \dots - 0.456309 \dots^2 = 0.0289805 \dots = \boxed{0.0290}$.

10 (d) $\sigma = 0.170236 \dots$, so the interval is from $0.286072 \dots$ to $0.626549 \dots$:

$$\int_{0.286072 \dots}^{0.626549 \dots} f(x) dx = 0.620012 \dots = \boxed{0.620}$$

11 (d) Below 0.75 kg nobody spends more than $25 \times 0.75 = \$18.75$. From 0.75 kg the price is $24x \leq 48$, so $x \leq 2$. The event is $0.5 \leq x \leq 2$:

$$\int_{0.5}^2 f(x) \, dx = 0.635294 \dots = \boxed{0.635}$$

Solving $25x \leq 48$ instead gives $x \leq 1.92$ and 0.601 — the discounted rate applies there.

11 (e) Two integrals, one for each rate:

$$\int_{0.5}^{0.75} 25x \, dx + \int_{0.75}^3 24x \, dx$$

$$= 1.51482 \dots + 39.4428 \dots = 40.9576 \dots$$

$$\text{so } \boxed{\$40.96}.$$

12 (a) $\int_3^9 (a(x-3)^3 + b(x-3)^2) \, dx = \left[\frac{a}{4}(x-3)^4 + \frac{b}{3}(x-3)^3 \right]_3^9 = 324a + 72b$, and that is 1.

$$\boxed{324a + 72b}$$

12 (b)(ii) $324a + 72b = 1$ and $6a + b = 0$:

$$\boxed{a = -\frac{1}{108}, b = \frac{1}{18}} \quad (-0.00926, 0.0556).$$

12 (c) The maximum of f on $[3, 9]$ is at $(7, 0.296296\dots)$: mode $\boxed{7}$. The area from 3 to 7 is $0.592592\dots > 0.5$, so the median is below 7; solving $\int_3^m f(x) dx = 0.5$ gives $m = 6.68563\dots = \boxed{6.69}$.

12 (d) Winning, $X > 8.52$, lies inside the condition $X > 8$:

$$\begin{aligned} P(X > 8.52 \mid X > 8) &= \frac{P(X > 8.52)}{P(X > 8)} \\ &= \frac{0.0344268\dots}{0.131944\dots} \\ &= 0.260919\dots \\ &= \boxed{0.261} \end{aligned}$$

The markscheme accepts 0.263 from a and b rounded to three figures.

Timer

(a)
$$E(X) = \int_0^2 \frac{6x}{\pi\sqrt{16-x^2}} dx = \frac{6}{\pi} \left[-\sqrt{16-x^2} \right]_0^2 = \frac{6}{\pi} (4 - \sqrt{12}) = \boxed{\frac{12}{\pi} (2 - \sqrt{3})}$$

(b) $P(X < 0.5) = \int_0^{0.5} f(x) \, dx =$
 $0.239358 \dots = \boxed{0.239}$